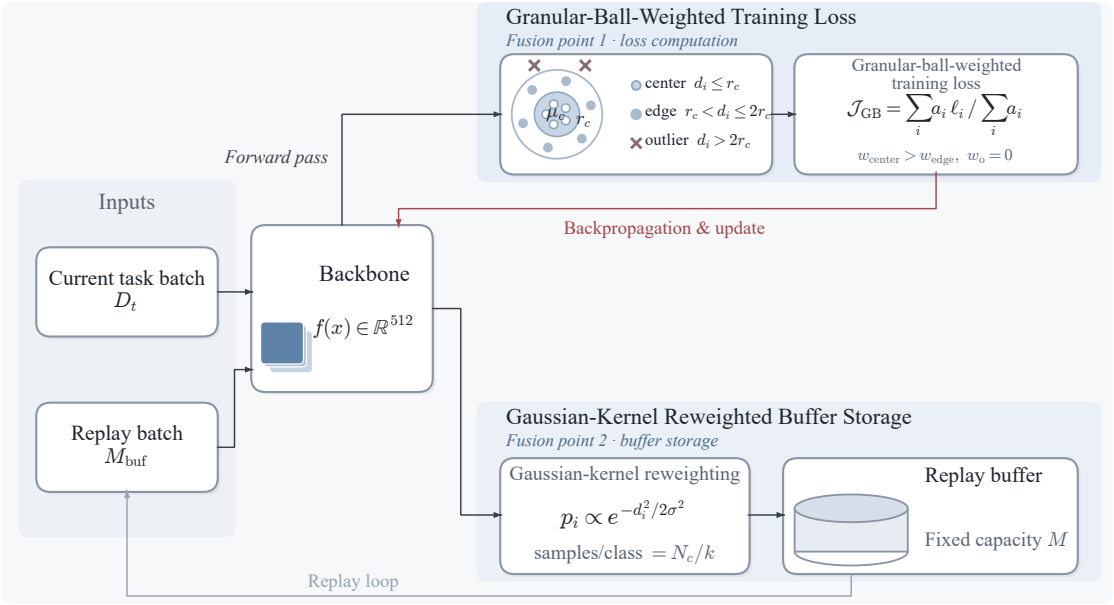

Granular-Ball-Weighted Training Loss
Fusion point 1 · loss computation
center $d_i \le r_c$
edge $r_c < d_i \le 2r_c$
outlier $d_i > 2r_c$
μ_c
r_c
Granular-ball-weighted training loss
$\mathcal{J}_{\mathrm{GB}} = \sum_i a_i \ell_i / \sum_i a_i$
$w_{\mathrm{center}} > w_{\mathrm{edge}},\ w_{\mathrm{o}} = 0$
Forward pass
Backpropagation & update
Inputs
Current task batch D_t
Backbone
$f(x) \in \mathbb{R}^{512}$
Replay batch M_{buf}
Gaussian-Kernel Reweighted Buffer Storage
Fusion point 2 · buffer storage
Gaussian-kernel reweighting
$p_i \propto e^{-d_i^2/2\sigma^2}$
samples/class $= N_c/k$
Replay buffer
Fixed capacity M
Replay loop